

### I Identification data

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| <b>Thesis title:</b>          | Distance estimate of autonomous helicopters by analysis of signal strength |
| <b>Author:</b>                | Fedor Strjapunin                                                           |
| <b>Type of thesis:</b>        | Bachelor                                                                   |
| <b>Faculty:</b>               | Faculty of Electrical Engineering (FEE)                                    |
| <b>Department:</b>            | Department of student                                                      |
| <b>Thesis reviewer:</b>       | Ing. David Fiedler, Ph.D.                                                  |
| <b>Reviewer's department:</b> | Cornell University, School of Civil and Environmental Engineering          |

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### II Evaluation

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| <b>Assignment</b> | <b>Extraordinarily challenging</b> |
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The assignment is demanding. It combines signal analysis with reproducible research software development, and either topic alone would be sufficient for a Bachelor thesis in my opinion.

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| <b>Fullfilment of assignment</b> | <b>Fulfilled with major objections</b> |
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The main goal of the assignment, "to analyse experimental data about location and signal strength from the flight of autonomous helicopters with the focus on reproducible research" was fullfied, except for the "reproducible research principles" part. The four sub-goals listed in the assignment also have been addressed, but only partially. Therefore, despite the significant amount of work done, I cannot say that the goals were fully achieved.

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| <b>Methodology</b> | <b>Correct</b> |
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The signal-analysis methodology seems correct and is supported by the results. However, the software implementation choices are questionable. One thesis goal is to "Discuss existing tools for reproducible research support and, if necessary, propose and implement a new system." I would expect Section 2.5 to justify the need for a new system and explain the software choices, but it does not.

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**Technical level****C - Good**

The thesis is technically sound, and the student seems to understand ranging methods well. However, Section 2.5.2 uses confusing terminology and seems out of place, while Sections 2.5.3 to 2.5.5 are shallow and incomplete. Another issue is the machine learning evaluation: although the test data are held out during model training, they are used for hyperparameter ( $k$ ) tuning, making the reported results more even questionable than the thesis admits. Finally, the limitations described in Section 6.2.1 are significant and not clearly justified.

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**Formal and language level, scope****B - Very good**

The mathematics and notation are mostly fine. The language is clear and understandable, with only a few minor typos. For future work, I suggest reducing the use of passive voice. The main weakness is organization: some paragraphs are misplaced, e.g., the second half of Section 2.2 is clearly not related work.

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**Sources and references****C - Good**

Overall, the number and quality of references seem good. What is missing is deeper integration of the references in the related work section. This section contains only a small number of references, and some claims are unsupported (e.g., Section 2.1).

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**Additional comments**

I would expect a thesis focused on reproducible research to include a public repository for both code and data. Additionally, the GUI images in Section 6 are unreadable.

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### III Overall evaluation and questions

Considering the overall challenging goals, I grade the thesis **B - Very good**.

Question 1: Even after the improvements achieved in this thesis, the absolute error still seems too high, in meters, which is a limitation acknowledged in the conclusion. Do you see further improvement options with the potential to reduce the error by at least an order of magnitude? Or is this a fundamental limitation of RSSI, and we should consider a different approach?

Question 2: One thesis goal was to follow reproducible research principles. However, the provided code targets a very narrow audience: Windows is the only supported platform, `uv` is required, and the implementation depends on a VSCode plugin. Since the rationale behind these choices is not described in the thesis, what strategy did you use to choose technologies for your implementation?

**Date:** June 1, 2026

**Signature:**